

CIRCULAR TIMBER DESIGN

Timber tracking strategy

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DSGN 2200 ASSESSMENT2 CIRCULAR TIMBER DESIGN

We Build the Future

BEST IDEAS \ BEST SOLUTION \ BEST RESULT

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2.1. Background Research

Why we should use recycled timber?

Eco-Friendly: One advantage of utilising recycled timber is its environmentally beneficial nature. There is no requirement for further deforestation for you to utilise trees, as you are solely repurposing discarded ones from others. If more individuals were to engage in the practice of timber reuse, it would potentially facilitate the recovery of forests.

Cost-effective: The cost of recycled timber is far lower than that of new timber, enabling individuals with limited budgets to purchase a greater quantity of materials (JUflooring, 2018)..

High-Quality: One of the primary advantages, which many individuals often overlook, is that recycled timber generally exhibits greater strength compared to newly sourced timber. Recycled timber exhibits a significantly reduced moisture content, approaching zero, owing to its prolonged drying period spanning several years. Consequently, this characteristic imparts enhanced durability and rigidity to the material (JUflooring, 2018).

How to start?

Timber is classified as a naturally occurring substance. However, waste timber typically encompasses additional components such as adhesives, coatings, and pigments, as well as diverse pollutants including timber preservatives and heavy metals. Furthermore, waste timber may also contain impurities such as glass, plastics, and metals. The presence of heterogeneity significantly complicates the recycling operations. As a result, the existing approaches for managing timber waste mostly revolve around landfilling, energy conversion, and material reclamation. The latter approach to recovery, despite its increased complexity, merits further development due to its reliance on recycling processes that generate new materials (Besserer, Troilo, Girods, Rogaume, & Brosse, 2021).

How much timber is wasted in Australia each year?

Determining the precise quantity of timber debris generated in New South Wales and Australia as a whole poses a challenging task. The following data presents figures from the year 2004: The state of New South Wales (NSW) was responsible for the disposal of approximately 315,000 tonnes of timber waste in landfills. A total of 131,000 metric tonnes were subjected to the process of recycling. Annually, the state of New South Wales (NSW) generated a total of 446,000 tonnes of timber waste. According to recent studies, it has been indicated that the annual quantity of timber waste generated constitutes around 6% of the total trash deposited in landfills throughout Australia. Although the numerical value may appear insignificant at first glance, this minute proportion signifies a substantial quantity of waste amounting to hundreds of thousands of metric tons, which is not being effectively repurposed or processed through recycling methods. (Liam, 2022)

Why some timber cannot be recycled

Painted, coated, and preservative-treated objects frequently include hazardous constituents, hence necessitating their disposal in landfills. In addition to this, glue is a very detrimental element to timber recycling, and most composite lumber uses Urea-formaldehyde (UF) glue, a very difficult glue to remove (Besserer, Troilo, Girods, Rogaume, & Brosse, 2021). Nevertheless, timber that has not undergone chemical treatment can often be subjected to the recycling process, e.g. reprocessed for sale as other wood or as a raw material such as fuel or fertiliser.

The presence of additional substances affixed to an otherwise valuable timber, hence diminishes its potential for reuse. Metal connectors, plasterboard, plastic wrapping, oils, dirt, sand, laminates (including some varieties of plywood), edge banding, glue, and resins are a few examples of the various impurities that can be discovered in the material (NSWepa, 2012).

How much of an impact does chemical treatment have on timber recycling?

Treated timber possesses a distinctive attribute that serves as a protective measure against several detrimental factors, including rot, decay, borers, and termites, safeguarding timber. The utilization of chemical preservatives, such as copper, chromium, and arsenic, serves to uphold the structural integrity of the timber

Nevertheless, it is important to acknowledge that treated timber does possess certain drawbacks and raises certain issues. The chemicals utilized for timber preservation might pose significant environmental risks if mishandled by individuals (Steffen, 2023).

What are the consequences of chemically treated timber that is not well disposed?

When opting to dispose of treated timber in a conventional landfill, the preservative chemicals contained within the timber are incinerated, resulting in their release into the atmosphere and subsequent seepage into the adjacent soil. The magnitude of pollution to this extent can have profound implications for our ecosystem. The infiltration of chemicals into the soil might subsequently contaminate our water sources, posing a potential threat of illness or disease to anybody who consumes it. Although it may appear to be a straightforward and convenient option for fueling your barbecue or timber heater, the act of burning pine or other types of lumber can pose significant hazards. The combustion of pine timber under high temperatures results in the emission of hazardous smoke due to the release of preservative compounds, posing a risk to respiratory health. The emission of smoke will rapidly compromise the air quality in the surrounding area, posing a threat to the well-being of individuals who inhale it (Steffen, 2023).

Some common chemical treatments

CCA

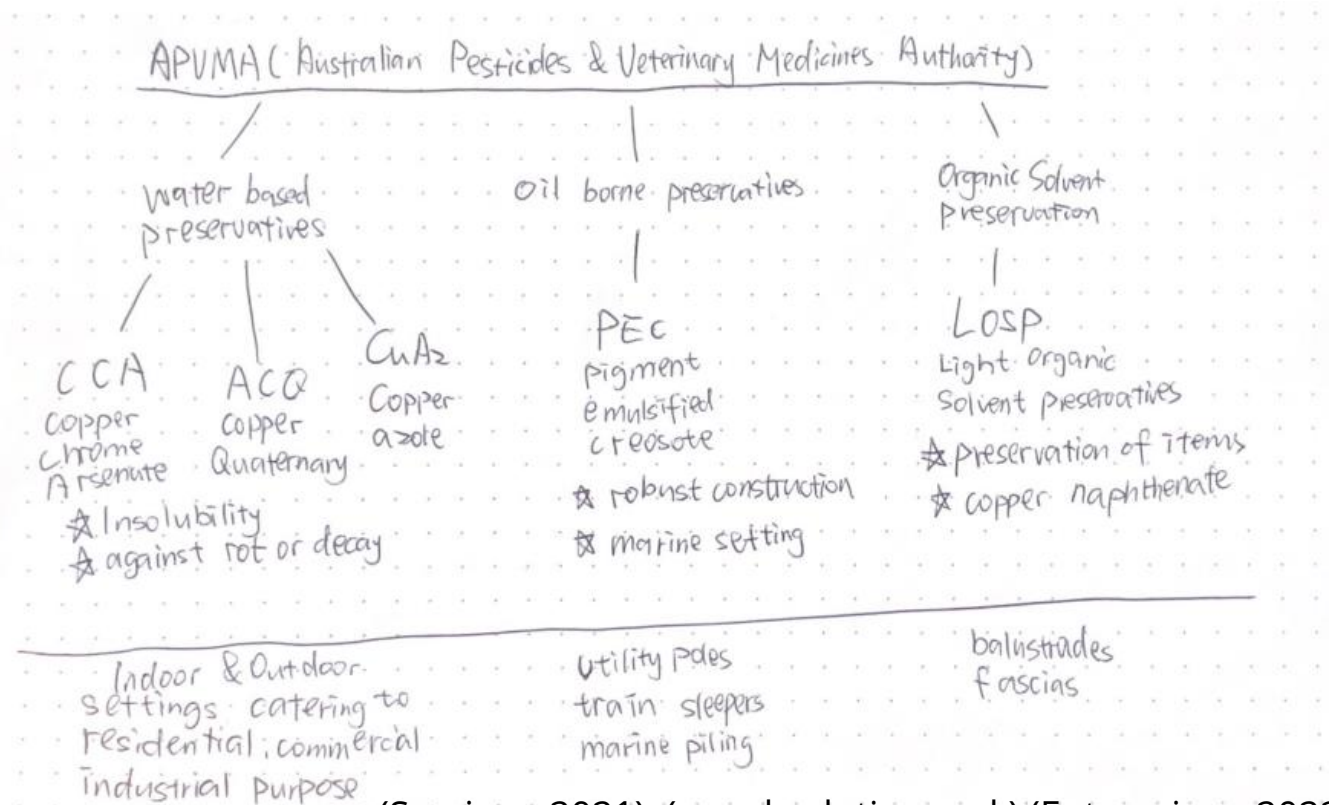
Copper chrome arsenate (CCA) treated timber refers to timber that has undergone a preservation process involving the application of a preservative solution comprising copper, chromium, and arsenic (Services, 2021).

PEC

Oil-based creosote/ pigment emulsified creosote (PECs) are employed outside as power poles and bridge piers. The pigment locks the primary HTC (high-temperature creosote) components into the timber so they don't flow onto the surface, especially in hot temperatures. The pigment keeps HTC's volatile lower boiling point fractions in the wood (wood solution, n.d.).

LOSP

The preservation method known as organo-biocide light organic solvent preservative (LOSP) utilizes azole fungicides to inhibit the decay of timber and permethrin insecticide to dissuade termites and other wood-boring insects (Enterprises, 2022).



(Services, 2021). (wood solution, n.d.) (Enterprises, 2022)

How should timber entering a landfill be?

Class (UK/Fin— Ger)	Chemical Composition	Origin	Panels Recycling Use	Energy Production
1 (A—AI)	Clean biomass	Packaging, solid wood offcuts	Yes	Yes (2910-A)
2 (B—AII)	Organohalogen and heavy metal thresholds	Building, waste from furniture	Yes	Yes(2910-B)
3 (C—AIII)	Other wood without hazardous substances	Waste from furniture, mixed waste	Yes or no depending on composition	Yes (2971 or 2771 depending on type)
4 (D—AIV)	Hazardous wastes	Exterior fittings (creosote, CCA, Cu- azole)	No	Yes (2770)

(Besserer, Troilo, Girods, Rogaume, & Brosse, 2021)

From the table above, we can see these classes primarily focus on the applications and corresponding regulatory constraints about the extraction of energy from panels or other sources. In France, the conventional categorization system typically comprises three classes, namely A, B, and C, which are defined as follows:

Class A pertains to clean or minimally treated timber. On the opposite side, class C is assigned to woods that possess hazardous compounds categorised as harmful. Primarily, these timber are characterized by the presence of heavy metals, such as CCA, or creosote. Class B is established based on its distinction as encompassing all waste materials that do not belong to either timber A or timber C. This course encompasses a wide range of materials that include wood-based additives, such as adhesives, coatings, finishing agents, veneers, and other related items. Hence, within this timber category, there is a minimal presence of polluted wood, characterised by only a few remnants of paints or varnishes, alongside a significant proportion of timber that is extensively contaminated with coatings, glues or even PVC edges(Besserer, Troilo, Girods, Rogaume, & Brosse, 2021).

2.2.Case study 01



(Queensland Government, 2023)

REUSE & RECYCLE

Reuse: Wood products that should be thrown away are put back into use by putting customised products that are no longer in production on display and letting people who want to use them again take them home.

Recycle: Sorting or processing replaced timber into raw

The Resilient Dwellings Fund in Queensland offers a voluntary buy-back programme for flood-prone dwellings. Despite these dwellings being destroyed owing to flood danger, the programme guarantees that their resources are not wasted. D & M Hose, for example, recycles construction materials and recovers salvageable components. The majority of the components are reused, including structural lumber, roof sheeting, doors, windows, and even unique aspects such as garage doors and roof fittings. They emphasise that the spectrum of recyclables is extensive, including items with unique forms and sizes like weatherboards and skirting boards. Beyond recovering raw materials, the effort also protects custom-made objects that are no longer in production. These unique components find homes with eager renovators, decreasing waste. The reclamation and repurposing process is ongoing, with approximately 360 buy-back proposals approved. This long-term strategy to rebuilding is driven by the Queensland and Australian governments, with assistance from DRFA. (Queensland Government, 2023)

2.2.Case study 02



(mike.wheeler, 2021)

REPURPOSE & REUSE

Repurpose: Recycled timber goes through screening, shredding, and agro-processing to be turned into fuel and fertiliser.

Reuse: Treated timber becomes artifacts or interior furnishings.materials.

Cleanaway is responsible for the collection of timber materials with the purpose of recycling them. In addition, they provide services for the disposal of treated timber. Cleanaway utilizes treated timber in the production of process-engineered fuel (PEF), in conjunction with other combustible materials, at our waste-to-energy facilities. According to their assertion, untreated timber (devoid of chemicals, paint, or varnish) can be utilized in agricultural applications as an organic recyclable material, contributing to soil enhancement and promoting plant development through its nutritional content. Chipped and shredded timbers possess a significant calorific value, rendering them suitable for utilization as furnace fuel stock. Additionally, these timbers can be combined and packaged into several compost mixtures. Cleanaway collaborated with its trade partners to engage in the advantageous repurposing of treated lumber, transforming it into various forms such as furniture, art, and customized pieces.(Cleanaway, 2023)

2.3.Case study finding

By using the background data from the previous survey and analyzing the three success cases, I believe recycle timber not only from the final stage of disposal, but also from the three stages of treating the timber, maintaining it in use and, most importantly, recycling it.

Chemical treatments are like a double-edged sword: chemically treated timber has a longer service life and is resistant to insects and decay, but is very labour-intensive to recycle and often requires large quantities of more chemical materials or special machinery to treat. In contrast, non-treated timber has a much-reduced service life and is at risk from insects and moisture, but the possibilities for recycling are greatly increased and the costs of recycling are relatively low.

So I think that timber recycling nowadays should find a balance between using or not using chemically treated timber. In the following, I will propose solutions to improve the recycling of wood, both from the point of view of producing it and recycling it.

3.1. Problem statement



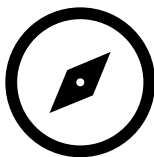
SELECTED PROBLEM

Because chemically treated timber is not well sorted or further processed, much of the timber that could have been recycled in is wasted.



INSIGHT

1. Global timber use is in short supply
2. Improper disposal of toxic timber leads to pollution of the environment
3. Loss of economic benefits from recycling
4. Cost of travelling to landfill
5. Waste of land resources by occupying landfill sites



DEFINE THE SCOPE

The main focus is on how to correctly sort and treat timber with the timber recycling sector, and to a lesser extent the sector that processes timber for the first time.



DESCRIBE PROBLEM CONTEXT

1. Awareness of the audience
2. Technical issue
3. The necessity of using non-recyclable components

3.2. Problem statement

How do we deal with toxic chemicals in timber, as timber treated with toxic chemicals and glues cannot be recycled, thus increasing the probability of the timber being reused?

Driver to change-->Quality/Treatment

After reviewing a great deal of information, I believe that one of the biggest problems holding back timber recycling is the polluting chemicals in the timber. For example, chemical treatments such as insect protection, and anti-rotting to increase lifespan; and chemical treatments such as painting, staining, and adhesives to increase aesthetics all affect the reuse and reprocessing of timber.

Therefore, I believe that if the chemicals in the timber are treated properly, the timber can be recycled better. This would not only reduce the environmental damage caused by improperly treated toxic timber but also reduce the potential threat of burning timber, such as global warming.

4.1. Vision Statement

Because timber treated with toxic chemicals and glues cannot be recycled, we want to use treatments and quality tests that will increase the probability that the timber will be reused.

Specific questions:

I would like to focus my subsequent research on timber that is difficult to recycle due to chemical treatments (preservative and insect control, etc.). This includes regulating the chemical treatment of timber before it is used and planning for the treatment and recycling of chemically treated timber after it has been scrapped. I think this is the most important and challenging part of the solution to the life of the timber loop.

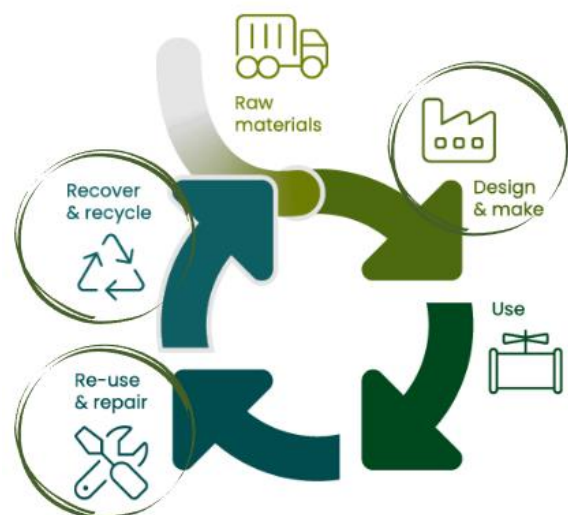
4.2. Vision Statement

What needs to change:

1. Timber is manufactured without rigorous regulation of chemical treatments (TPAA, n.d.).
2. The provenance and preservation of timber cannot be accurately traced.
3. The timber that enters the dump is not subjected to sorting procedures, hence hindering the identification of recyclable timber.
4. The incineration of timber without implementing a process of segregation into distinct treatments, hence resulting in the emission of pollutants that negatively impact the environment.
5. A significant volume of hazardous timber material accumulates and occupies the designated landfill area.

Why do these problems deserve to be changed?

These problems prevent the circular life of timber from closing the loop, which is not conducive to a circular economy. Improving the treatment of timber from the three perspectives of production, restoration and recycling is a very effective way to realize a circular economy. Improving the treatment of timber improves the performance and longevity of timber and brings multiple



4.2. Vision Statement

Envisaged outcome

Through the implementation of many envisioned methodologies, the objective I aspire to attain is the establishment of a timber recycling system characterised by a self-renewing process. This process commences with the tree and culminates with the return of the timber to its end-of-life state, wherein it is transformed into nutrients that facilitate the nourishment of subsequent trees. In conjunction with regulatory and standards assessments, the manufacturing and recycling processes of timber are expected to undergo increased stringency, hence enhancing the prospects of timber recycling throughout time. Implementing the three fundamental principles of the circular economy:



**Plan to reduce
waste and
pollution**



**Circulate goods &
materials**



restore nature

5.1. Circular design strategies

Cyclic design strategies: The design method employed in this study aims to effectively address the two distinct stages of the cyclic life of wood. Firstly, the process of sorting will be implemented, followed by the utilization of all recycled wood, which will be further elaborated upon in the subsequent stages of the concept design.

In the proposed recycling plan, the objective is to optimize the use of timber resources through the implementation of the three Rs (Repurpose, Reuse, and Recycle) approach, ensuring that each component is effectively utilized. I aim to select materials that prioritize safety and environmental sustainability from their origin, with the ultimate objective of minimizing pollution at every stage of the process.

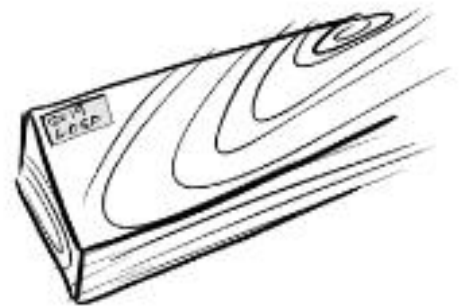
To establish a cyclical life cycle, the dangerous compounds included in the recycled timber are eliminated, and afterward, the timber is crushed into timber chips. These timber chips are then utilized as a form of fertilizer, providing nourishment to facilitate the growth of new trees. Furthermore, from a technological standpoint, it is anticipated that the Iron exchange membrane technique(Booking, 2007) can facilitate the re-refinement of CCA heavy metals into copper. This copper can then be utilized in various applications, including the creation of copper molds for lettering during the initial production phase, as later mentioned.

6.1. Concept Design

Change 1: Implement stringent regulations to effectively manage chemical treatments that have adverse environmental impacts or pose challenges in their disposal, such as chromated copper arsenate (CCA), during the initial stages of timber manufacturing. Mitigate the potential risks associated with detrimental additives by employing alternate options.



Change 2: Carve into each cut piece of timber at the time of production, using a drill bit, the process by which the timber has been treated. For example, 2019, LOSP treated, up to twelve years of life.

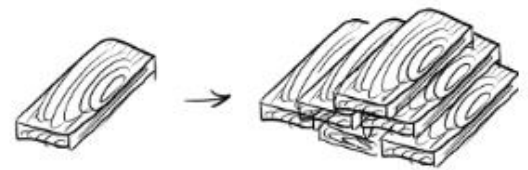


Change 3: Upon disposal in a landfill, the timber undergoes two distinct grading procedures. The first, known as visual grading, entails an assessment of the timber's labels, softness, hardness, and potential suitability for recycling. The second, referred to as machine grading, involves an examination of the timber's internal resilience, condition, and presence of toxic or non-recyclable substances.

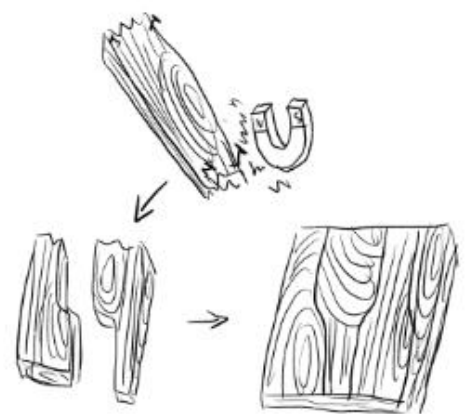


6.1. Concept Design

Change 4 (Reuse) : The timber that is disposed of in landfills has been categorised into three distinct classes. The reclassification of A grade timber is warranted to reflect its status as either "in good condition, ready for use" or "requiring cutting/processing for utilisation." The portion that is in a ready-to-use state can be instantly forwarded to construction businesses for potential reuse, while the piece that requires processing will be held in conjunction with the B-grade timber for subsequent steps.

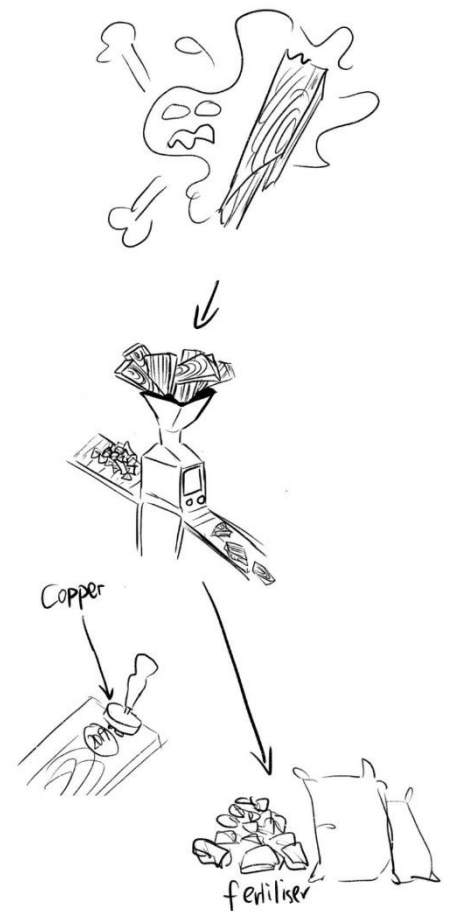


Change 5 (Repurpose) : In order to render B-grade lumber suitable for recycling, it is necessary to eliminate a minor fraction of non-recyclable components, including adhesive substances, paint, and small quantities of nails. This procedure entails the utilisation of metal detectors and substantial magnets to extract sizable metal objects, such as nails, followed by the identification of timber sections requiring cutting using computer analysis. The cutting process is executed with utmost efficiency to minimise wastage. The timber that has been divided into smaller pieces is subsequently combined with the other components of Part A in order to produce fibreboards or wooden crafts that may be sold again, so achieving the objective of repurposing.



6.1. Concept Design

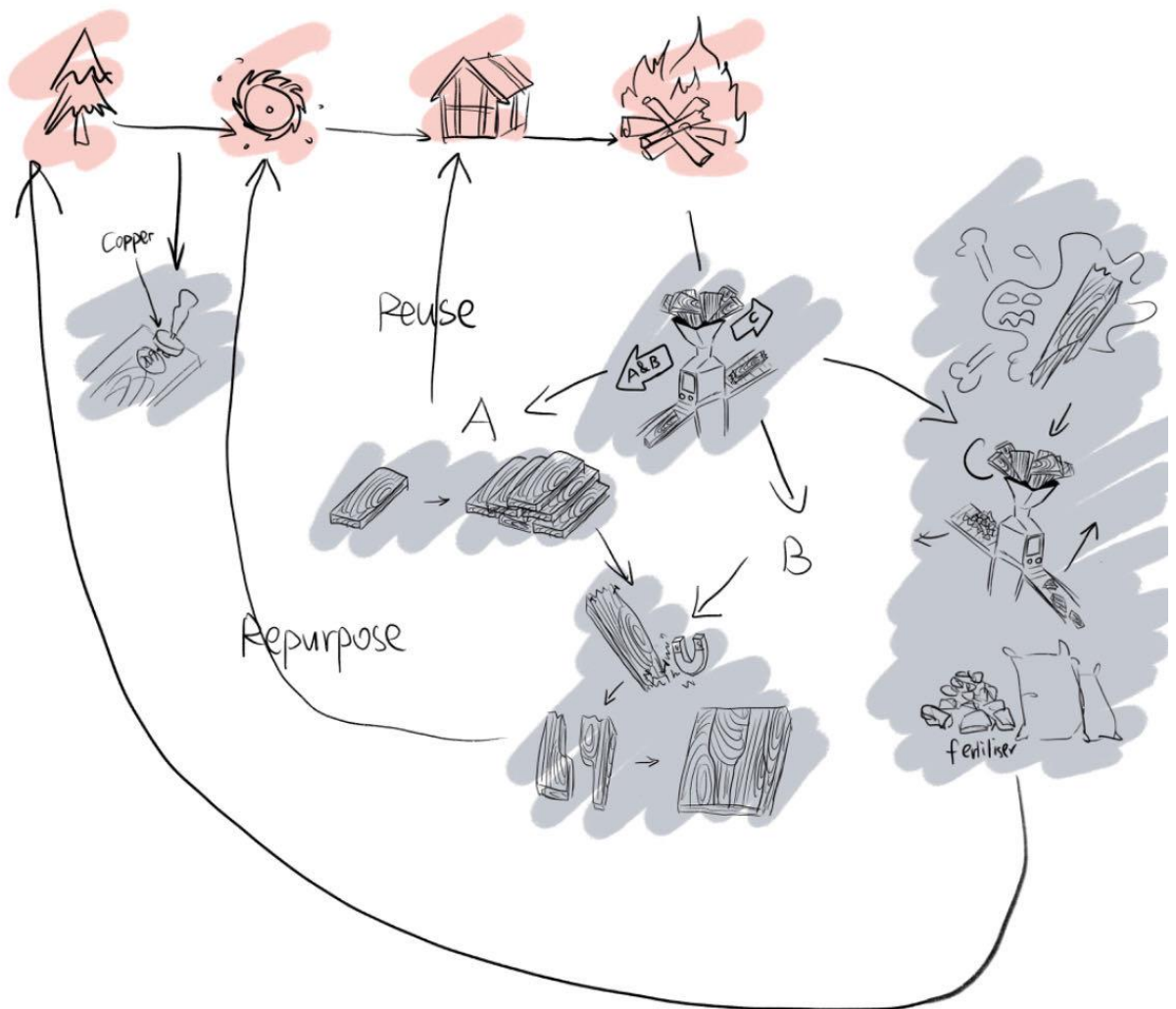
Change 6 (Recycle) : Wood classified as C is known to present considerable difficulties due to its high content of poisonous compounds. These substances pose environmental risks if the timber is either burned immediately or left unattended for extended periods. Consequently, it is strongly advised to exercise caution while dealing with this particular timber category. One such approach involves the utilisation of iron exchange membrane technology (Booking, 2007) for the purpose of extracting heavy metals present in timber treated with chromated copper arsenate (CCA). The copper separated by the machine can be used as the raw material for the copper branding in the second change. Following an extensive examination of the treated timber through a series of rigorous quality assessments, I propose that with the elimination of heavy metals from the C-grade wood, there exists a viable opportunity to engage in the recycling of both the fuel and raw materials employed in the treatment process.



6.2. Concept Design

Challenges:

1. Ability of the solution to be implemented in practice, monitoring norms and transparency.
2. Consumption of energy and cost of resources
3. The potential cost of sorting timber and the benefits of timber recycling are not comparable.
4. Damage to employees when handling toxic chemicals.



A2

Timber Treaking Strategy

BEST IDEAS \ BEST SOLUTION \ BEST RESULT

7.1.Introduction of A2

To enhance the efficacy of the wood recovery plan, I made the strategic decision to prioritise the implementation of a comprehensive programme aimed at identifying and locating reliable sources of timber-related data. The objective was to mitigate the intricacy associated with the sorting of construction timber following demolition, enhance the likelihood of timber recyclability, and minimise environmental pollution resulting from inappropriate disposal of timber containing hazardous chemicals.



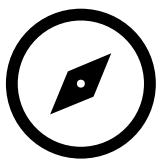
SELECTED PROBLEM

The inability to trace the origin of the timber used at Queenslander resulted in an inability to maximise the recovery of usable timber after demolition.



How to develop & expected

Record the treatment of timber and guide recycling simultaneously by labelling the timber and database collation.



DEFINE THE SCOPE

Specializing in Queenslander construction demolition and recycling for Queensland timber situations and solutions.



TARGET AUDIENCE

Environmental agencies,
Timber producers and processors,
Builders,
Demolition companies,
Consumers.

7.2. Vision Statement

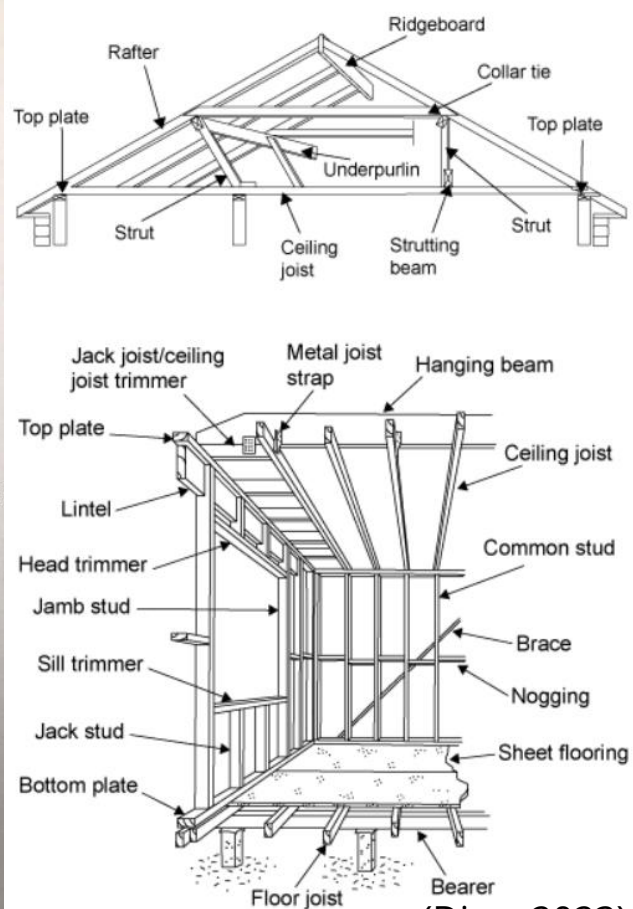
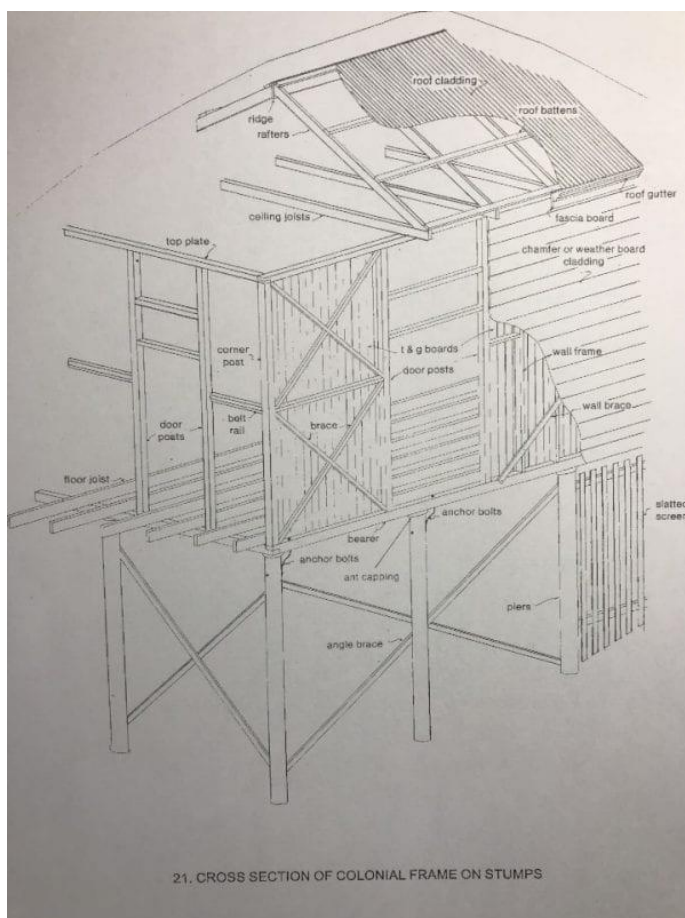
The efficient identification of recyclable wood is hindered by the challenges associated with tracing its origin and treatment methods. In order to enhance the probability of wood reuse, My solution is to implement a labeling system for the wood and establish a relational database linkage.

8.1.Futher Research

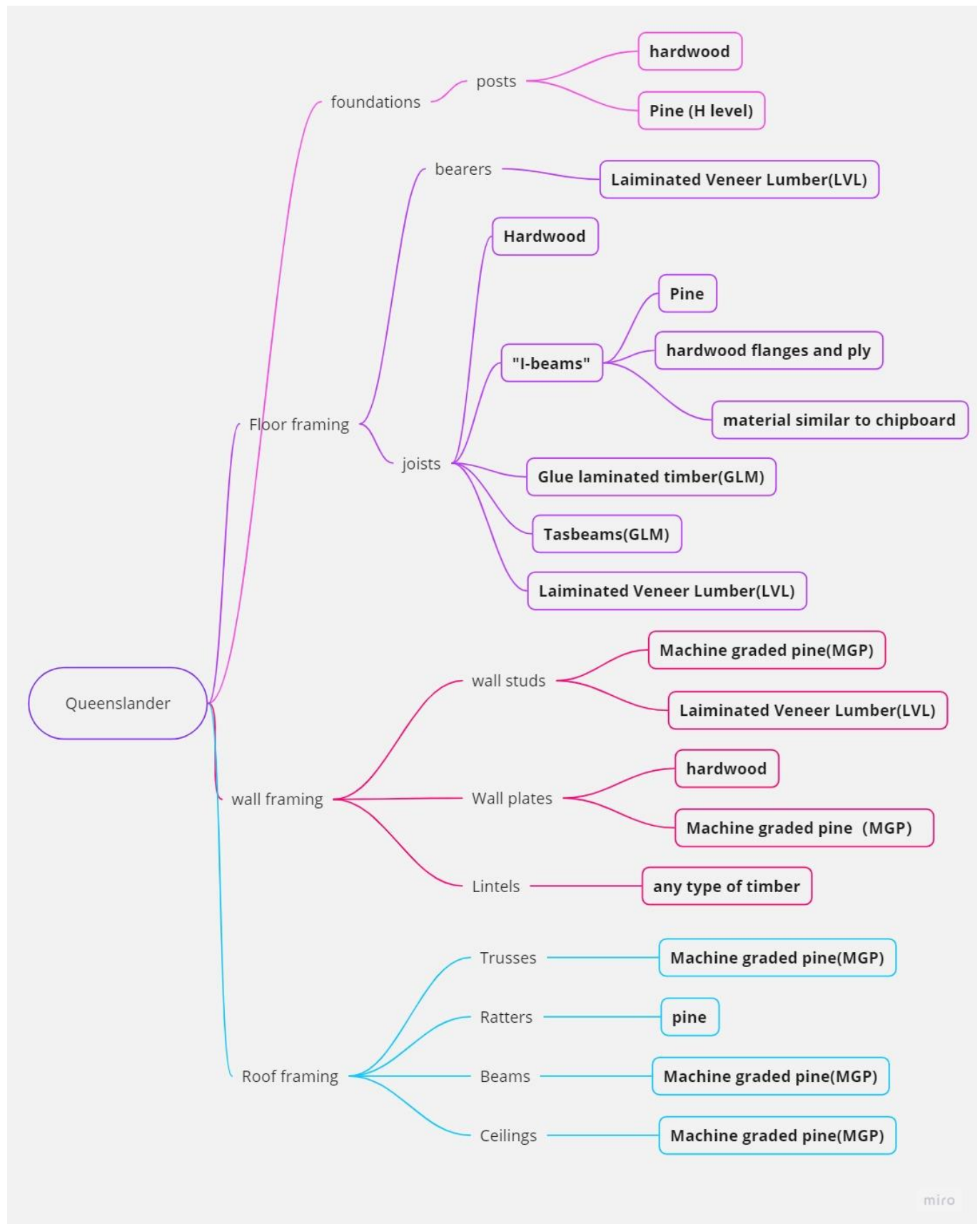
What timber is used in the construction of Queenslander ?

Built in the late 1800s and early 1900s, Queenslander architecture is exclusive to Queensland and some areas of New South Wales. Most were built using local wood that was easily accessible, because of transportable, and reasonably priced (Peter, 2020).

The fact that Queenslanders were constructed on tall, wide-diameter timber stumps provided for excellent circulation to the subfloor, which helped keep the home somewhat cool throughout the summer (Peter, 2020).



(Peter, 2020).



(Grant, 2005)

Machine graded pine(MGP)

MGP Timber – MGP10 vs MGP12 vs MGP15

Suitable to judge the strength of joists and bearers, MGP (machine graded pine) assesses compressive and structural strength of the pine timber, as well as the stiffness. The more stress the pine timber can withstand, the more MGP it'll have.

There are three ratings available for MGP:

MGP10 = F7 Highly versatile all-round timber Minimum requirement for building	MGP12 = F8 Higher strength than MGP10 Suitable for frames, walls, floors joists and supports	MGP15 = F11 Higher strength than MGP12 Suitable for frames, walls, floor joists and supports
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(Narangba Timbers, 2021).

MGP Structural Pine is a very suitable option for a wide range of construction purposes, owing to its multitude of advantages that effectively address the needs of builders, contractors, and homeowners alike. The lumber in question is produced with a focus on strength, durability, and versatility, rendering it well-suited for applications such as framing, flooring, and other structural components within residential and commercial construction projects. The MGP Structural Pine undergoes treatment to effectively deter termites and prevent rot, so safeguarding one's investment against potential harm. Moreover, this product serves as an environmentally sustainable option in comparison to alternative construction materials. The utilization of MGP Structural Pine offers builders the assurance of constructing with a sense of security, as they select a dependable and enduring material that encompasses elements of visual appeal, robustness, and environmental consciousness within a unified entity(canterbury timber, n.d.).

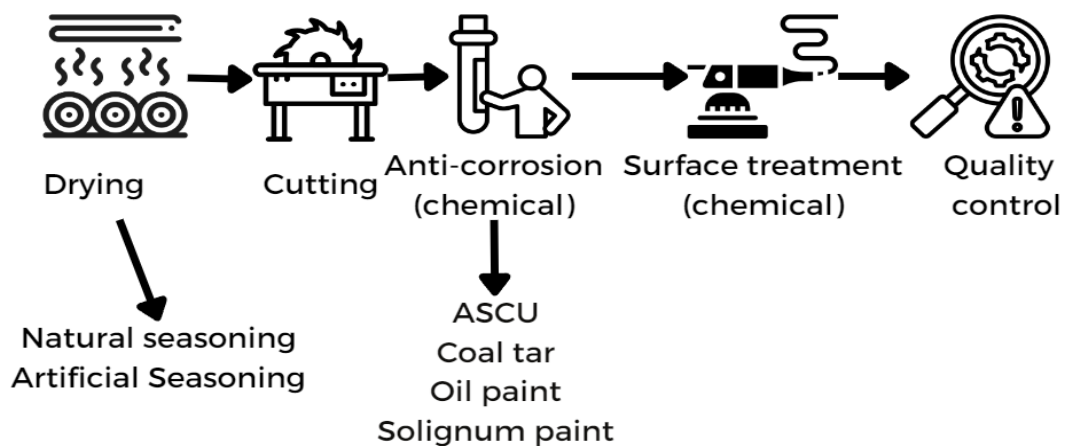
Laminated Veneer Lumber(LVL)

Laminated Veneer Lumber (LVL) is a constituent of a collection of goods known as structural composite lumber (SCL). These SCL products are manufactured by assembling dried and graded wood veneers, strands, or flakes in layers, which are then bonded together using a moisture-resistant adhesive. The resulting composite material is formed into sizable blocks referred to as billets(naturally wood, n.d.).

What happened in timber used in Queenslander now?

1. require constant upkeep
2. Lightweight wood is more susceptible to termite infestations and offers inadequate insulation against intense heat.
3. Every ten to fifteen years, wood must be repainted to account for the swelling and shrinking caused by intense heat.
4. When building a home out of wood, wood rot is a regular problem.
5. Tin roofs are more resilient than tile during tropical storms, but they intensify the intense heat.
6. Even though they are built with termite mitigation in mind, they must be constantly monitored to keep pests away (Hohenadel, 2022).

what are the characteristics of these timbers (how have they been treated)



(The Constructor, 2018)

What information needs to be passed on

- Wood species and description
- Hazard class
- Timber Durability Classes
- Strength
- Preservative name
- Penetration value
- Retention value
- The wood preserver's name and identification number.

Hazard level illustration

The level of biological hazard that timber will be subjected to in a given application is determined by the hazard levels. The different degrees indicate the chemical treatment applied to the wood to strengthen its resistance against termites, rot, and fungal development. On a scale ranging from H6, the highest level, to H1, the lowest level, the danger level is graded (Narangba Timbers, 2021).

Hazard level						
	attack by insect	attack by termites	wet	near ground	water	ocean
H ₁	✓					
H ₂	✓	✓				
H ₃	✓	✓	✓			
H ₄	✓	✓	✓	✓		
H ₅	✓	✓	✓	✓	✓	
H ₆	✓	✓	✓	✓	✓	✓

(Narangba Timbers, 2021).

What is the purpose of tracking and recording timber?



Sustainable
Forestry
Management



Classification of
timber recycling



Regulation of
Chemical
Treatments

To whom is this information to be communicated



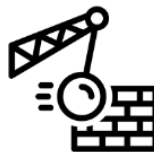
Timber suppliers
and producers



Real estate
companies and
individual
consumers



Builders and
construction
workers



Demolition sector



Government and
environmental
supervisors



Recyclers (target
audience after
dismantling)

How the timber will be used after demolition

1. Timber that is in a satisfactory state can be resold or directly recycled by the homeowner or property company for utilization in the construction of a new dwelling.
2. Wood that is in a state of fair condition has the potential to undergo processing and be transformed into wood composite materials, including chipboard, fibreboard, particleboard, and other similar products. Alternatively, it can be utilized for the production of furniture, flooring, and wooden embellishments.
3. Certain types of wood that are unsuitable for use as construction timber due to chemical treatments can still be utilized for the production of biomass energy. These wood materials can be processed into wood pellets or chips, which can then be employed as a source of biomass energy, such as biomass fuel. This particular type of energy possesses the capability to be utilized for many purposes such as heating, power generation, and fulfilling other energy requirements(Stewardship & Strategies, 2017).

What is the outcome of passing information

The primary objective is to examine the recurring life cycle of wood and optimize its qualities in order to enhance its durability through a comprehensive analysis of its origins.

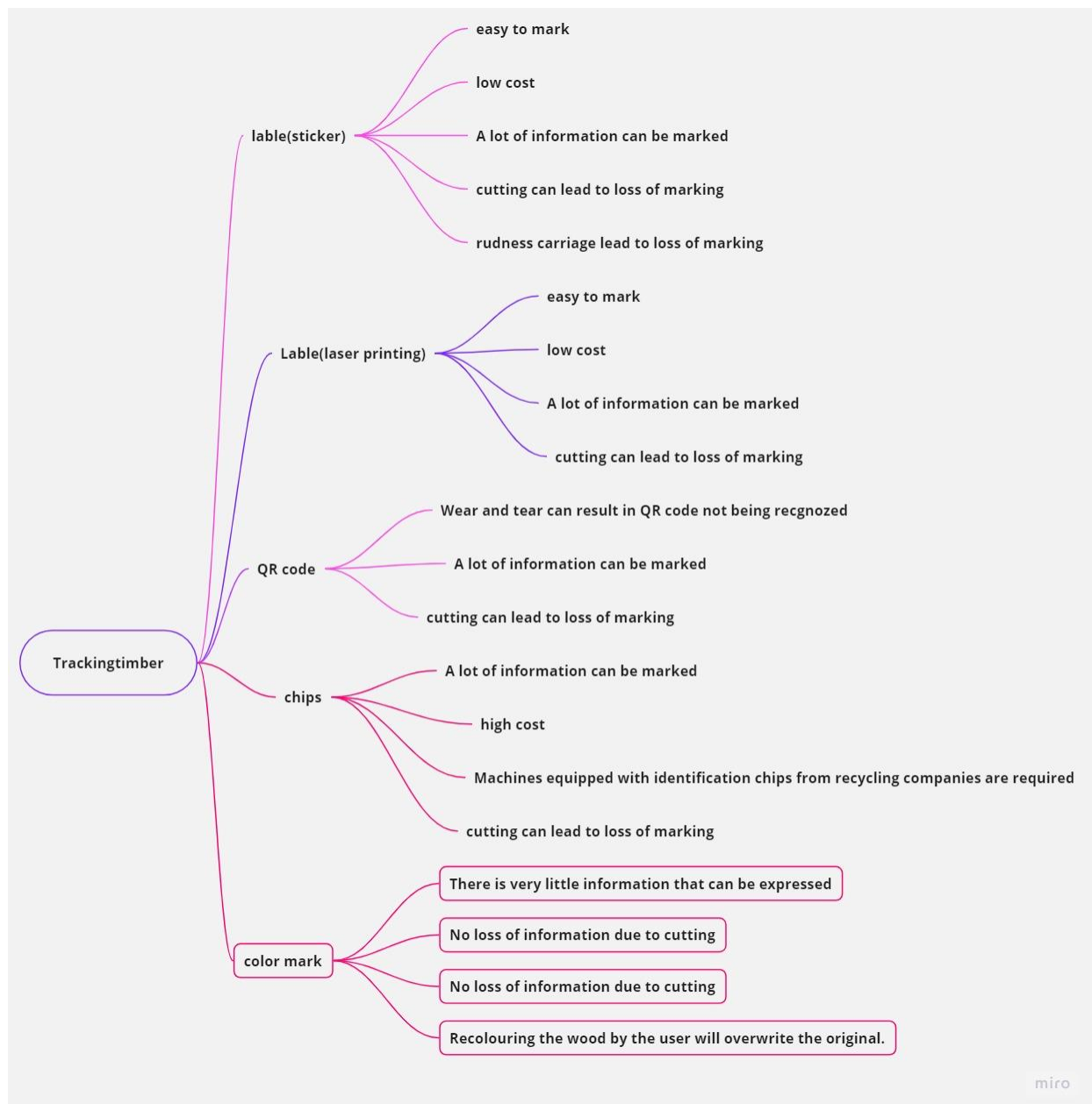
Understanding how the wood is processed and how it should be treated after it is dismantled will not only increase the re-utilisation of the wood but also avoid environmental pollution caused by improper treatment.

9.1. Prototype(color track)

Aim:

Trace the timber physically, visually represent the chemical treatments and sources the timber has received, and clarify what to do with it and how to classify it.

Brainstorming:



miro

Brainstorming result: spraying colour is the most appropriate way to trace the origin and treatment of timber.

Problems still need to be fixed:

Q1:How can I make the colour and wood blend better together?



The first method:

The wood is stained by immersing it in coloured water. This method will absolutely prevent meter damage due to uncut or wear and tear in transport, but it will change the wood's original colour, reducing its value dramatically. (Not everyone wants a blue home).



The second method:

For the second way, paint the timber's sides (the parts that won't be seen). This fixes the issue of colored wood. If you use this method, you still run the risk of losing marks, and because it is nailed together, you might hard to diffrenciate when it is being taken apart.



The third method:

I did some study and found a pigment that changes color depending on the temperature. At room temperature, it is clear, but in a certain temperature range, it turns color. You can get the right color code if you paint this pigment on all sides of the wood and heat it when you recycle it. I'll explain more on the next page

Thermochromic Material



(iSuoChem®, n.d.)

Thermochromic pigments are tiny particles that change color over and over again when the temperature changes. Organic color-making systems with unique chemical structures are called electron-transferring organic molecules. At a certain temperature, electron transfer changes the molecular structure of the organic substance, which is what makes the color change. This color-changing substance is not only colorful, but it can also change from "colored ==> colorless" and "colorless ==> colored." This is something that heavy metal salt complex type and liquid crystal type temp-sensitive color-changing substances do not have. Reversible thermochromic pigment, which is also called thermochromic pigment, thermochromic powder, or thermochromic powder, is a reversible thermochromic material that is packed into tiny particles. The color's particles are round balls about 2 to 7 microns across (a micron is a thousandth of a millimeter). Inside is the substance that changes color, and on the outside is a clear layer that is about 0.2 to 0.5 microns thick and can't melt or dissolve. This layer keeps other chemicals from wearing away the substance that changes color. Because of this, it is very important not to damage this layer of shell while it is being used (Bianse Feng, n.d.).

Q2: Does it cover the colour code if the user wants to re-colour the timber?

The answer is yes, thermochromic pigments are simply a coating that covers the surface of the timber. In order to address this solution, we have strict rules for tenants not to allow large areas of timber to be colored. This is because firstly, we are unable to detect whether the paint is hazardous or not, which makes it more difficult to recycle; and secondly, if there are more than two surfaces, the thermochromic pigments may be completely covered, which will certainly make it more difficult to distinguish between the colors when it comes to recycling.

The solution we offer is: firstly, the user needs to buy pigments that meet the specifications (they will not contaminate the environment when recycled). Secondly, the user needs to contact the builder on the website before applying the pigments to a large area, contact the relevant person to assess the composition of the pigments and spray the surface with thermochromic pigments after applying the pigments.



9.2. Prototype(Database)

I believe that relational databases can be used in the delivery of information, and by creating integrated databases, data from different areas can be linked together for easy querying and management(Harrington, 2016).

ID	Species	Purpose	HazardLeve	Durability	Preservative	Strength	Company	ColorRepresentative
001	Pine(H3)	RoofFrame	H3	D1	ASCU	F14	A company	Red
002	MGP15	WallFrame	H3	D1	ASCU	F12	B company	Yellow
003	LVL	FloorFrame	H3	D1	SolignumPaint	F14	C company	Blue
004	Pine(H3)	Posts1	H3	D2	ASCU	F14	D company	green
005	Pine(H4)	Posts2	H4	D1	ASCU	F27	E company	Black

ID is primary key
Unique identifier

ColorRepresentative	treatment	purpose	FutherPaint
Red	Cut&Treatment	Reuse	Orange
Yellow	Cut&Treatment	Reuse	Orange
Blue	Grind up	Fertiliser	NA
Green	Burn	IndustrialFuel	NA
Black	landfill	NA	NA

Color is foreign key
Connect with other table's primary key

Color is primary key

Table 1

Material Table (Queenslander)								
ID	Species	Purpose	HazardLevel	Durability	Preservative	Strength	Company	ColorRepresentative
001	Pine(H3)	RoofFrame	H3	D1	ASCU	F14	A company	Red
002	MGP15	WallFrame	H3	D1	ASCU	F12	B company	Yellow
003	LVL	FloorFrame	H3	D1	SolignumPaint	F14	C company	Blue
004	Pine(H3)	Posts1	H3	D2	ASCU	F14	D company	green
005	Pine(H4)	Posts2	H4	D1	ASCU	F27	E company	Black

The presented diagram illustrates the timber utilised in the construction of a Queenslander (assumed), along with its corresponding condition and other pertinent information. The colour scheme employed in the diagram serves to denote the many aspects being represented. The chart in question serves as a digital backup, originally created by the builder and possessed by several stakeholders including the homeowner (developer), builder, demolisher, and environmental organization. The primary objective of this table is to establish a correlation between intricate information and corresponding colors, hence facilitating subsequent retrieval and referencing.

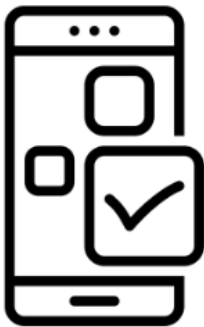
Table 2

post-dismantling wood treatment			
ColorRepresentative	treatment	purpose	FutherPaint
Red	Cut&Treatment	Reuse	Orange
Yellow	Cut&Treatment	Reuse	Orange
Blue	Grind up	Fertiliser	NA
Green	Burn	IndustrialFuel	NA
Black	landfill	NA	NA

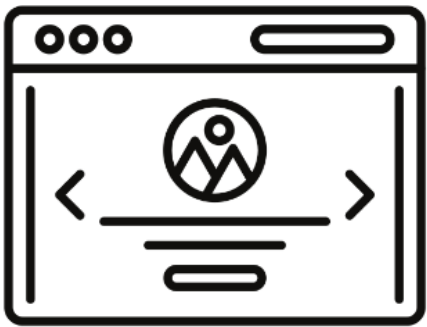
In contrast to its predecessor, the present table is designed to cater to those engaged in the demolition of residential structures. These individuals may possess limited knowledge of the characteristics and classifications of various types of wood, hindering their ability to effectively and efficiently choose the appropriate disposal destination for each specific wood material, whether a recycling facility or a landfill site. Utilizing the provided table and implementing color distinction will enable the demolition crew to effectively discern the appropriate course of action for each color category. This approach offers a notable level of convenience and efficiency.

How does each user access the database?

Because of the wide variety of treatments that wood undergoes, it is essential to make the database easy and fast for every user. And, for easy communication and unified management, a powerful app or website will be the best medium.



APP



Website

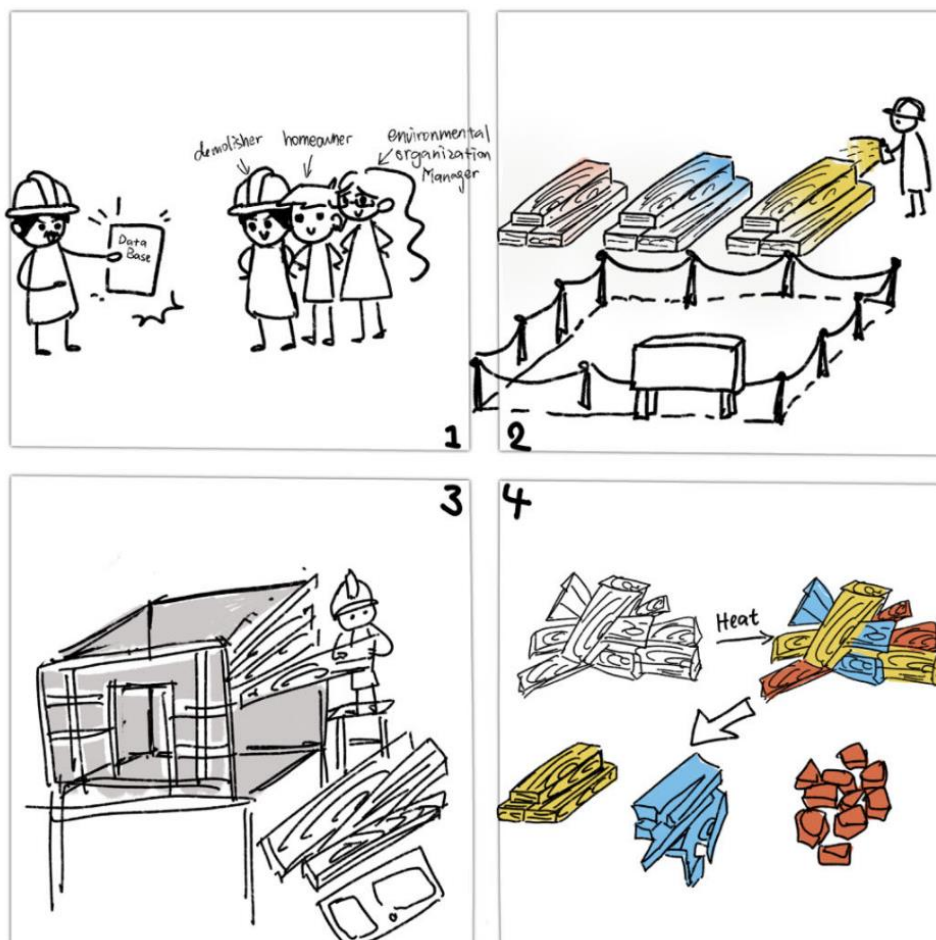


Cloud database

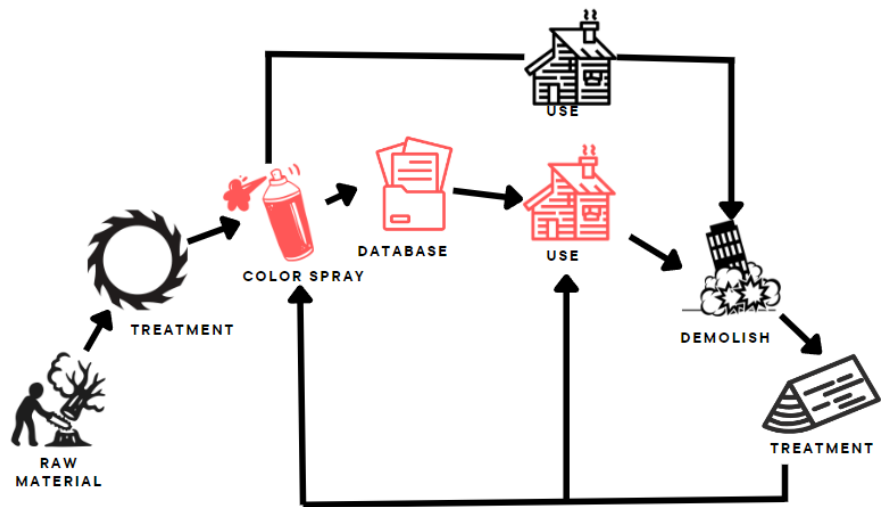
10.1. Concept Design

A brief overview of the design process

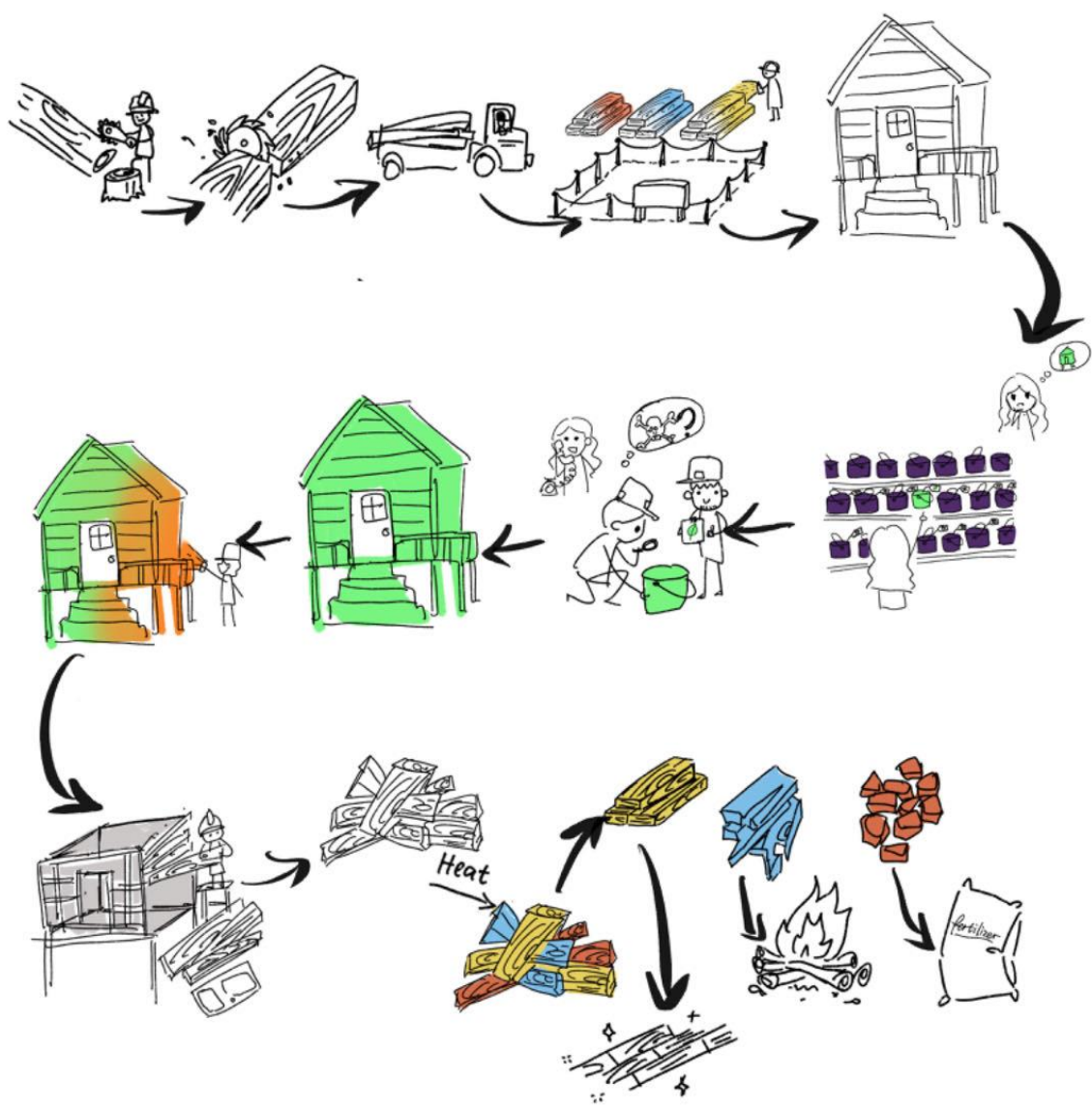
The concurrent utilization of thermochromic materials and their corresponding databases enables the dissemination of information to many stakeholders including manufacturers, constructors, end-users, environmental organizations, dismantlers, and recycling entities. This facilitates the collective awareness of the handling of wood and the subsequent strategies employed. The attainment of transparency and correctness in information is ensured throughout the entirety of the process, hence optimizing the utilization of wood recycling.



Flow chart



Lifecycle of Queenslander



10.1. Industry & stakeholder feedback analysis

PROBLEM OF waste

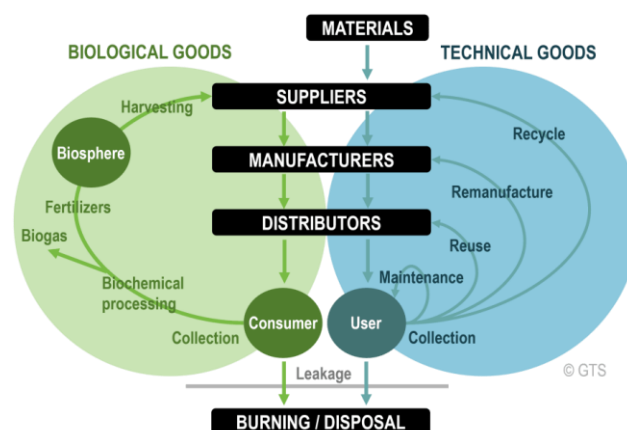
Implementing this strategy creates some new waste, such as the cost of painting and labor costs. To solve this problem I have devised two future solutions:

The first is to soak the wood instead of spraying it with paint.

Since thermochromic pigments can be dissolved in water and have the same effect, the wood can be immersed in a huge pool of thermochromic pigments (Wang et al., 2018). The goal is to recycle energy and technology. Suppose in the future the wood needs to be painted in a new color. In that case, there is no need to worry about overwriting the traces of the original because the thermochromic pigments are already incorporated into the wood, and removing the other pigments from the surface is only necessary.

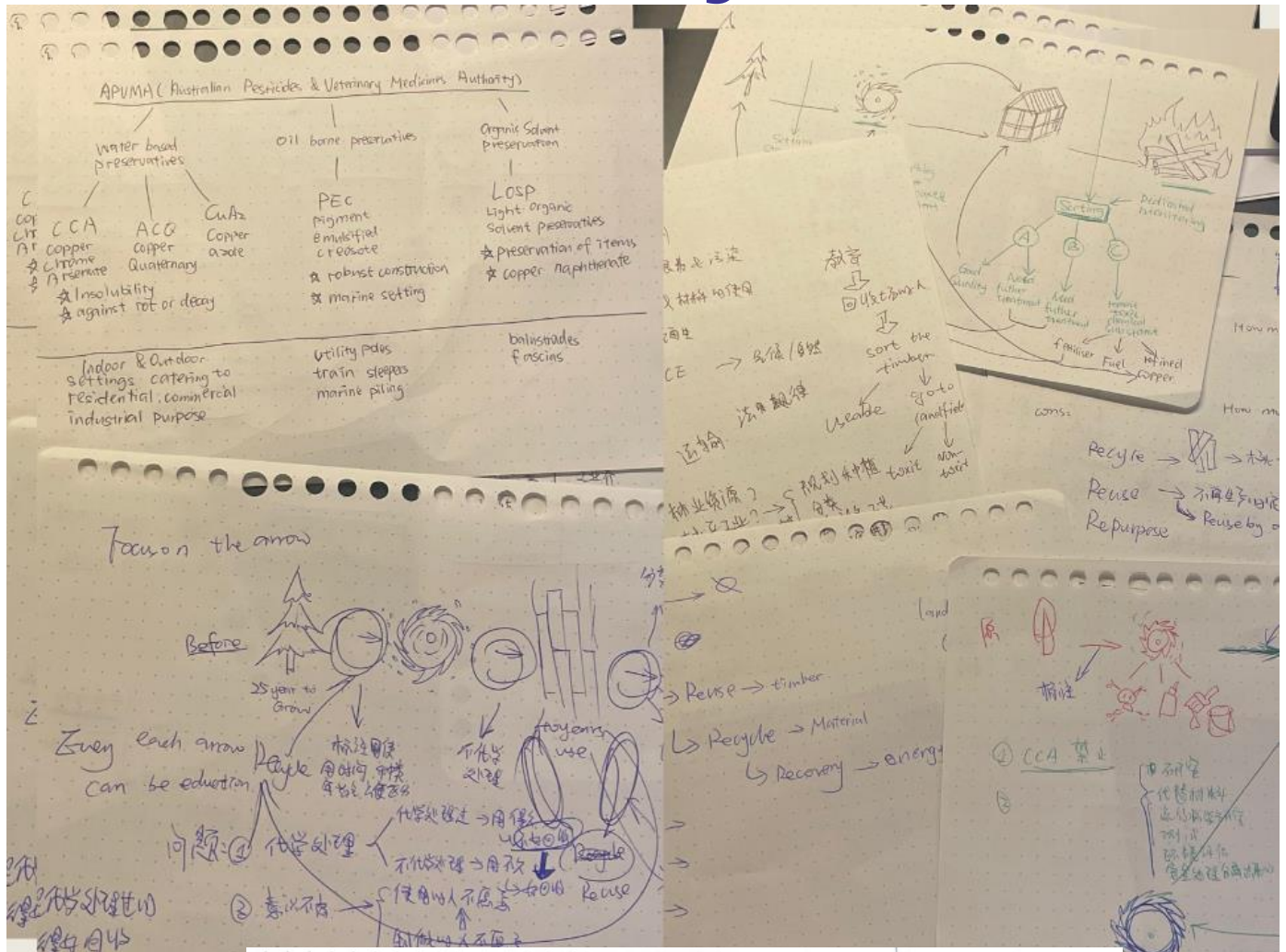
Secondly, it reduces the need for repetitive painting.

The previous design strategy was that the user would need a professional to re-spray the thermochromic pigment when re-coloring any wood, which would incur a significant portion of the cost. After some rethinking, I think it is possible to make it so that part of the wood needs to be re-treated, such as wall paneling and roofing, but smaller or less recyclable wood such as steps, door frames, or stakes can be left alone at the discretion of the user. Users can search the database (APP) and find out what parts do not need to be repainted.

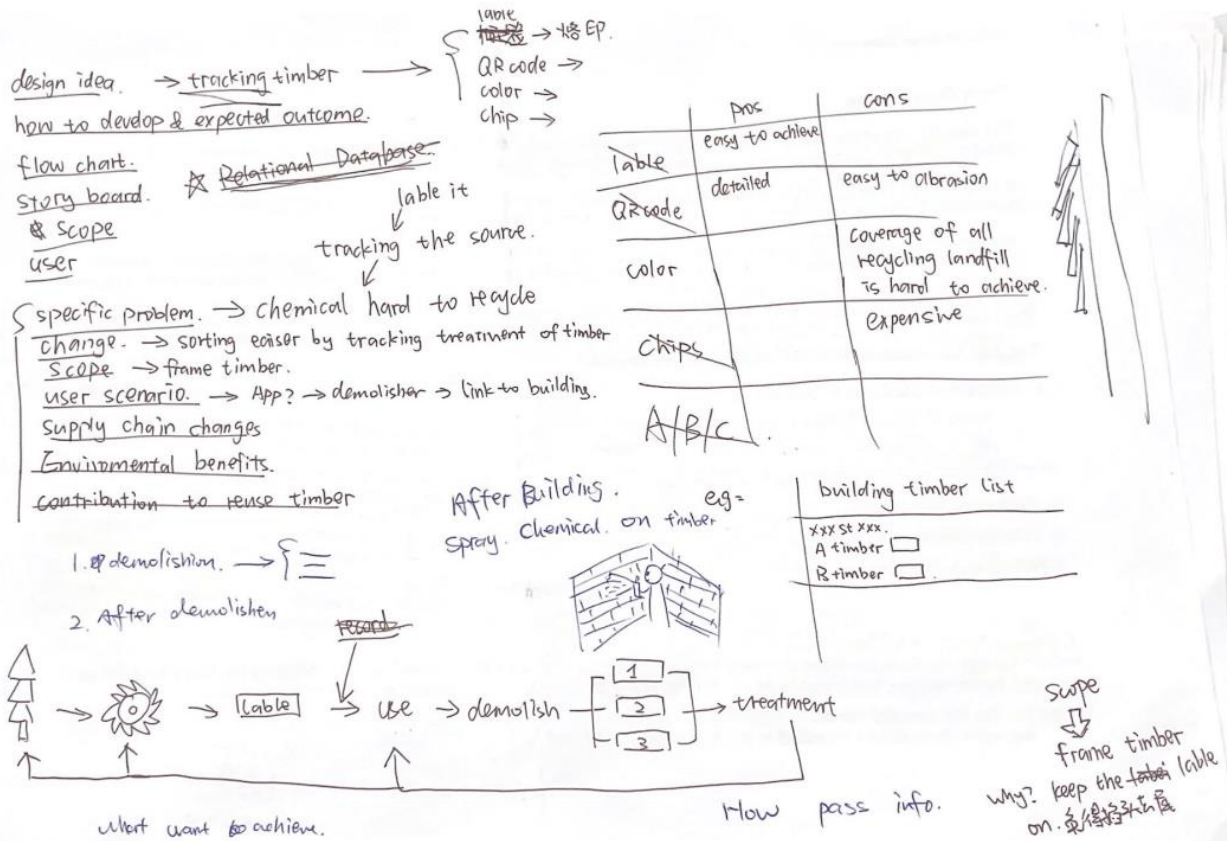


(The Geography of Transport Systems, n.d.)

11.1. Reflective journal(A1)



11.2. Reflective journal(A2)



COLOR

Temperature sensation.

Thermochromic Ink

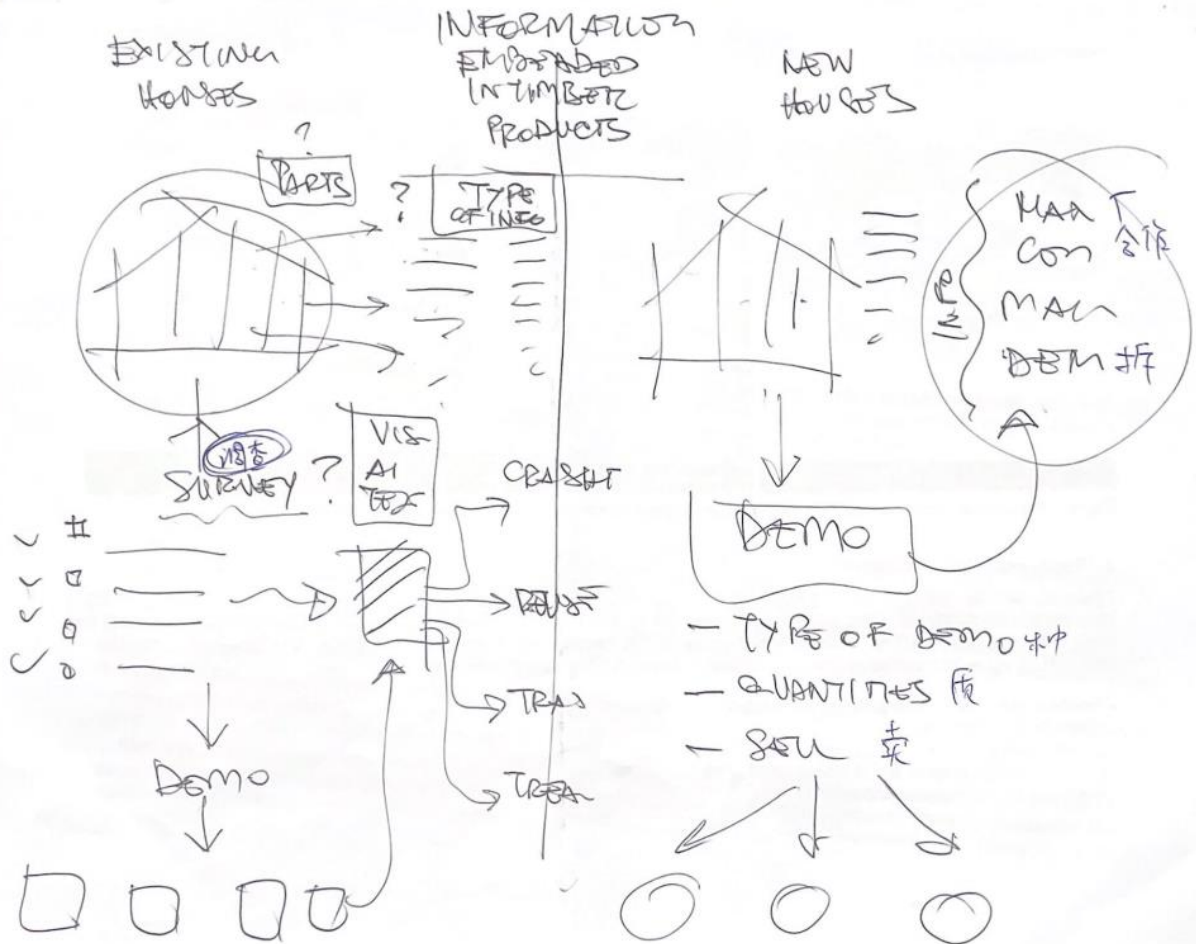
PROS:

- Many color to choose.
- Easily notice the difference in color and easily distinguish the origin of the wood.
- It's ~~not~~ impossible to tell the difference because the wood has been cut.

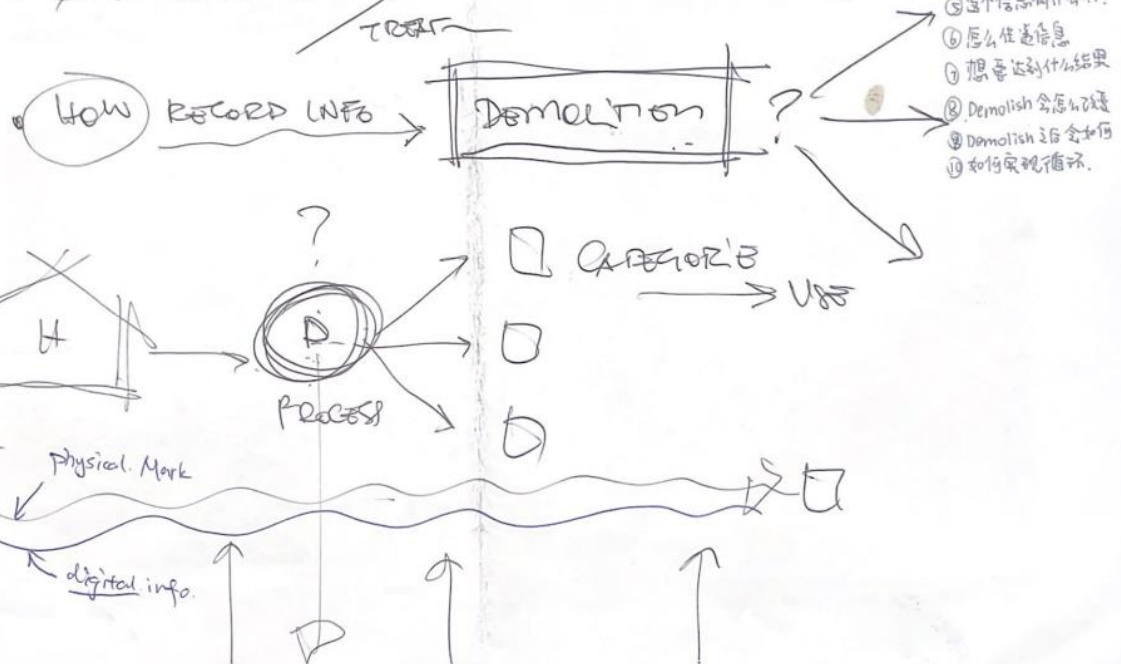
CONS:

- Uncertainty of temperature at which the wood is exposed
- extremely high or low temperature cause chemical reaction in discolored material resulting in an irreversible color

Spray after building



- PARTS OF HOUSE (TIMBER COMPONENTS)
- ≠ TIMBER TYPE



1. 有哪些木材会被用? Queen slonder
2. 它们有什么特性(经过什么处理)
3. 要传递什么信息?
4. 传递信息给谁?
5. 这个信息有什么用.
6. 怎么传递信息
7. 想要达到什么结果
8. Demolish 会怎么改变
9. Demolish 指令如何
10. 如何实现循环.

unique identifier

1-D

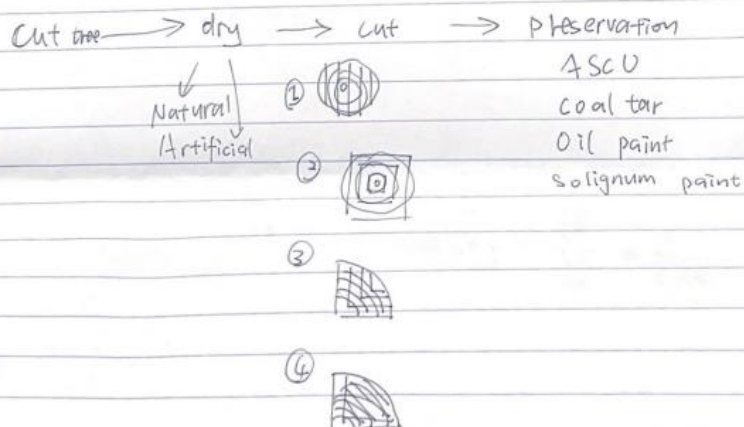
用途	屋顶	墙面	地板	地基B	地基A
品种	Pine (H3)	MGP 15	LVL	Pine (H3)	Pine (H4)
危害度	H3	H3	H3	H3	H4
耐久	D1	D1	D1	D2	D4
防腐	ASCU	ASCU	Solignum paint	ASCU	
硬度	F14	F12	F14	F27	
Company	A公司	B公司	C公司	D公司	

2-D

颜色

红	地基A
黄	地基B
蓝	地板
绿	墙面
橙	屋顶

Hazard level		attack by insect	attack by termites	wet weather	near ground	water	ocean
H1		✓					
H2		✓	✓				
H3		✓	✓	✓	✓		
H4		✓	✓	✓	✓	✓	
H5		✓	✓	✓	✓	✓	✓
H6		✓	✓	✓	✓	✓	✓



posts — { hardwood (F14 or F27)
pine C"H" level

Floor Framing — Any timber (dry)

Beaters — LVL

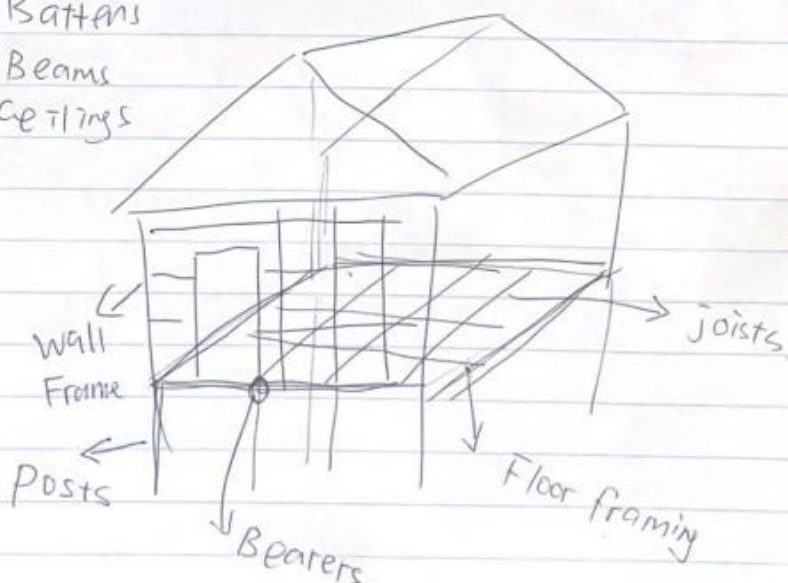
Joists {
• hardwood
• "I-Beams" → { pine
hardwood flanges.
ply
material similar to chipboard.
glue laminated timber.
• "GLM" Torsbeams.
• Laminated Veneer Lumber (LVL)

Wall Frame { plates, noggings, lintels & bracing
wall studs
↓ → MGP. machine graded pine
Hard wood.

→ bearers
lintels
joints
rafter.

Roof Framing { Trusses
Rafters
Battens
Beams
Ceilings

屋顶
墙面
窗户
结构. 框架
楼梯
栏杆



CIRCULAR TIMBER DESIGN

CHEN ZHONG 47208566

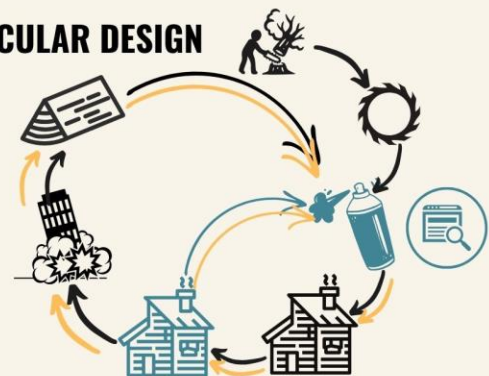
Color Code Tracking Strategy



PROBLEM TATEMENT

Since information on the timber itself used in Queenslander cannot be ascertained at the time of recycling, and therefore the use of the timber cannot be maximized, how can the origin and treatment of the timber be traced?

CIRCULAR DESIGN



DRIVE TO
CHANGE
Treatment

THERMOCHROMIC PIGMENT



RELATIONAL DATABASE

ID	Species	Purpose	HazardLevel	Durability	Preservative	Strength	Company	ColorRepresentative
001 Pine(H2)	RoofFrame	H3	D1	ASCU	F24	A company	Red	
002 MGPS15	WallFrame	H3	D1	ASCU	F12	B company	Yellow	
003 LVL	FloorFrame	H3	D1	ASCU	F14	C company	Blue	
004 Pine(H2)	Post1	H3	D2	ASCU	F14	D company	Green	
005 Pine(H4)	Post2	H4	D1	ASCU	F27	E company	Black	

ID is primary key
Unique identifier

Color is foreign key
Connect with other table's prime



SOLUTION

Target Audience



Builders & construction workers



Demolition sector



Government & environmental supervisors



Recyclers (target audience after dismantling)



Timber suppliers & producers



Real estate companies & individual consumers

What info need to be pass

- wood species and description
- hazard class
- Timber Durability Classes
- Strength
- preservative name
- penetration value
- retention value
- the wood preserver's name and identification number.



Color Code strategy

1 Problem statement

- drive to change.

2 Solution.

① color code.

② relational database.



5 what info need to be passed.



6 story board.



Q still need be fixed

① updates

③ Link between color & database.

④ How pass info. (App / website)

⑤ Further idea.



Feedback.

① Focus on the label. (Tracking)

↓
Prototype.

④ Talk to demolition people.

② Testing Function

③ how it works.

④ physical reaction

Elevator Pitch.

电梯法则 → 30秒

① 3句之内

② 语言惊人

③ 短小精悍

④ 提炼观点

value. when
who (the users) where
What
why (why user use it)
How it works
action item.
what's next.
name of object.

★ Tone

Information

Direct, clear, concise. - eye contact.
- trust.
- interact with audience.

body language

confidence.

knowing your pitch

Hook.

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COLOR CODE TRACKING STRATEGY
CHEN ZHONG 47208566
DSGN2200 CIRCULAR TIMBER FRAME

